



OPEN
Compute Project

Usage Guide for Ethernet NIC

v1.0.0

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1 Version History

Date	Version	Author	Description
11/4/2025	1.0.0	Jeff Hilland	Initial usage guide and profile contribution



2 Introduction

This document describes the manageability tasks that can be performed on a platform which is conformant to the Ethernet NIC v1.0 Profile.

The document shows how the Redfish interface is used to accomplish these tasks. This includes examples of the interaction across the Redfish interface.

The document references the OCP Profile as the normative form of the requirements.



3 Compliance to Open Compute Project Tenets

3.1 Openness

The usage guide and profile were developed on a public OCP GitHub repository. An open-source validator exists for validating conformance of an implementation to the profile.

3.2 Efficiency

A common manageability interface reduces the amount of client-side code required to be written, and enables the remote client to be used with various conformant products.

3.3 Impact

Manageability standards enable the interchange of interoperable products. The Ethernet NIC profile prescribes the common manageability behavior of Ethernet NICs.

3.4 Scale

Redfish is one of the scalable interface for managing platforms out-of-band used on OCP platforms.

3.5 Sustainability

No impact.



4 Management Tasks

The following table lists the management tasks and their requirements for the Ethernet NIC.

OCP's out-of-band platform manageability interface is based on Redfish. Redfish is an international recognized standard, specified by [Redfish API Specification](#) [1] and the [Redfish Data Model Specification](#) [2].

As a Redfish interface, the Ethernet NIC requirements on the Redfish model elements has been normatively specified in a corresponding [OCP Ethernet NIC Profile v1.0](#) [4]. The syntax of the profile document is specified in the [Redfish Interoperability Profile Specification](#) [5]. Issues with the profile can be filed on the [OCP Profile repository](#).

The profile document can be read by the [Redfish Interop Validator](#) [3] to test the conformance of an implementation. The validator will auto-generate tests, execute the tests, and create a test report. Issues with the validator can be filed on the Redfish discussion board ([redfishforum.com](#)).

Use Case	Management Task	Requirement
NIC Configuration	Get NIC information	Mandatory
	Get device function information	Mandatory
	Get port information	Mandatory
	Get PCIe information	Mandatory
	Get PCIe function information	Mandatory
	Set NIC information	Mandatory
	Set device function information	Mandatory
	Set port information	Mandatory
	Set NIC settings to default	Mandatory
NIC Hardware	Get FRU Information	Mandatory
System NIC	Get system NIC information	Mandatory
Telemetry	Get NIC metrics	Mandatory
	Get device function metrics	Mandatory
	Get port metrics	Mandatory

Use Case	Management Task	Requirement
Firmware Update	Get firmware version	Mandatory
	Update firmware	Mandatory



5 Redfish Model for NICs

The Redfish model use cases can be better understood, if some aspects of the Redfish model are comprehend first. The [OCP Ethernet NIC Profile Reference Guide v1.0](#) [7] describes the resources which are part of this profile. The [OCP Ethernet NIC mockup](#) [8] shows a mockup of the resource hierarchy in an example implementation.

Redfish models a managed entity in terms of its physical and logical aspects:

- The physical aspect is modeled via the Chassis resource. This includes NetworkAdapter, NetworkDeviceFunction, Ports, PCIeDevice and PCIeFunction. Some of these resources have subordinate metrics resources as well.
- The logical aspect is modeled via the ComputerSystem. This is done using EthernetInterface and represents the System view, or logical or OS view, of the NIC.

The relationship between the above resources are specified by the Links property.

- On the Chassis resource, there are links to the resources that together describe the NIC. This includes NetworkAdapter, NetworkDeviceFunction, Ports, PCIeDevice and PCIeFunction.
- On the ComputerSystem resource, there are links to the NetworkDeviceFunction that describes the NIC.

The following diagram helps to show the relationships between these Redfish resources:

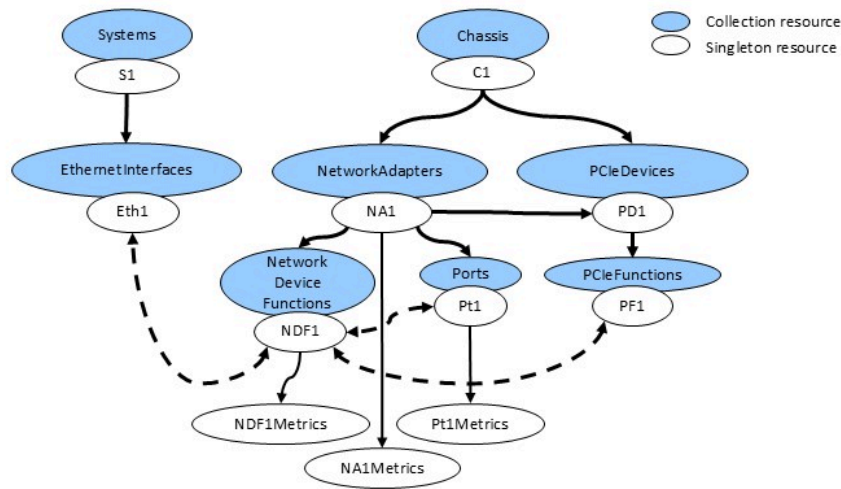


Figure 1: Redfish NIC Hierarchy

Many NICs are multi-function devices. It is important to discover the state of the physical card, the number of Ports and their information, how many functions are on the card and PCIe (tm) information for the card and the functions. This is begun by finding the NetworkAdapter under the Chassis.

5.1 Setting Information

Setting live network information is not required. If the implementation supports setting any of the properties at the next system reset, those settings are exposed through a `Settings` object in each resource.

5.2 Metrics

There are metrics objects on the NetworkAdapter, NetworkDeviceFunction and the Port. The objects contain properties of metrics which may change quickly over time.

6 Use Cases

The section below describes the interaction with the Redfish interface to fulfill the management tasks. In the example responses, a fragment of the full response or resource may be shown.

6.1 Get NIC Information

The following illustrates how to get the NetworkAdapter resource.

```
GET /redfish/v1/Chassis/1/NetworkAdapter/DE07A000
```

The response message contains the following fragment.

```
{
  "@odata.type": "#NetworkAdapter.v1_9_0.NetworkAdapter",
  "Id": "DE07A000",
  "Name": "Network Adapter",
  "Manufacturer": "Contoso",
  "Model": "Contoso 2",
  "SKU": "Contoso 2 function adapter",
  "SerialNumber": "LMNOP4279",
  "PartNumber": "ABCDEFGG2",
  "Status": {
    "Health": "OK",
    "State": "Enabled"
  },
  "Ports": {
    "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports"
  },
  "NetworkDeviceFunctions": {
    "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/NetworkDeviceFunctions"
  },
  "Metrics": {
    "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Metrics"
  },
  "Controllers": [
    {
      "FirmwarePackageVersion": "229.1.123.0",
      "Links": {
        "PCIeDevices": [
          {
            "@odata.id": "/redfish/v1/Chassis/1/PCIeDevices/DE07A000"
          }
        ]
      }
    }
  ]
}
```



```

    ],
    "ControllerCapabilities": {
      "NetworkPortCount": 4,
      "NetworkDeviceFunctionCount": 16,
      "DataCenterBridging": {
        "Capable": true
      },
    },
    "NPAR": {
      "NparCapable": true,
      "NparEnabled": true
    },
    "VirtualizationOffload": {
      "SRIOV": {
        "SRIOVVEPACapable": true
      },
      "VirtualFunction": {
        "DeviceMaxCount": 256,
        "MinAssignmentGroupSize": 8,
        "NetworkPortMaxCount": 256
      }
    },
    "PCIEInterface": {
      "LanesInUse": 8,
      "MaxLanes": 16,
      "MaxPCIEType": "Gen4",
      "PCIEType": "Gen4"
    }
  },
  "LLDPEnabled": true,
  "Actions": {
    "#NetworkAdapter.ResetSettingsToDefault": {
      "target": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Actions/NetworkAdapter.ResetSettingsToDefault",
      "@Redfish.OperationApplyTimeSupport": {
        "@odata.type": "#Settings.v1_3_3.OperationApplyTimeSupport",
        "SupportedValues": [
          "OnReset"
        ]
      }
    }
  },
  "@odata.context": "/redfish/v1/$metadata#NetworkAdapter.NetworkAdapter",
  "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000",
  "@odata.etag": "W/\"4DFAAF27\"",
  "@Redfish.Settings": {
    "@odata.type": "#Settings.v1_3_3.Settings",
    "SettingsObject": {
      "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Settings"
    }
  }
}

```



```

    },
    "SupportedApplyTimes": [
        "OnReset"
    ]
}
}

```

6.2 Get Device Function Information

Also found in the Network Adapter are the NetworkDeviceFunctions and the Ports, so retrieving those are needed. First we will get the NetworkDeviceFunction. Note that the NetDevFuncType must equal Ethernet for this profile to apply.

```
GET /redfish/v1/Chassis/1/NetworkAdapter/DE07A000/NetworkDeviceFunctions/1
```

The response message contains the following fragment.

```

{
  "@odata.type": "#NetworkDeviceFunction.v1_8_0.NetworkDeviceFunction",
  "Id": "1",
  "Name": "Network Device Function 1",
  "Status": {
    "Health": "OK",
    "State": "Enabled"
  },
  "Metrics": {
    "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/NetworkDeviceFunctions/1/Metrics"
  },
  "NetDevFuncType": "Ethernet",
  "DeviceEnabled": true,
  "NetDevFuncCapabilities": [
    "Ethernet"
  ],
  "Ethernet": {
    "MACAddress": "9c:dc:71:c3:bb:0a",
    "PermanentMACAddress": "9c:dc:71:c3:bb:0a",
    "MTUSizeMaximum": 9600
  },
  "BootMode": "PXE",
  "VirtualFunctionsEnabled": true,
  "MaxVirtualFunctions": 8,
  "AssignablePhysicalNetworkPorts": [
    {

```

```

        "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports/1"
    }
],
"Links": {
    "PCIeFunction": {
        "@odata.id": "/redfish/v1/Chassis/1/PCIeDevices/DE07A000/PCIeFunctions/1"
    },
    "PhysicalNetworkPortAssignment": {
        "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports/1"
    },
    "EthernetInterfaces": [
        {
            "@odata.id": "/redfish/v1/Systems/1/EthernetInterfaces/5"
        }
    ]
},
"@odata.context": "/redfish/v1/$metadata#NetworkDeviceFunction.NetworkDeviceFunction",
"@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/NetworkDeviceFunctions/1",
"@odata.etag": "W/\"80212509\""
}

```

6.3 Get Port Information

This is an example of retrieving a Port object

```
GET /redfish/v1/Chassis/1/NetworkAdapter/DE07A000/Ports/1
```

The response message contains the following fragment.

```

{
    "@odata.type": "#Port.v1_6_0.Port",
    "Id": "1",
    "Name": "Ethernet Port 1",
    "Status": {
        "Health": "OK",
        "State": "Enabled"
    },
    "Metrics": {
        "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports/1/Metrics"
    },
    "PortId": "1",
    "PortProtocol": "Ethernet",
    "PortType": "BidirectionalPort",
    "Enabled": true,

```



```

"Ethernet": {
  "SupportedEthernetCapabilities": [
    "WakeOnLAN"
  ],
  "AssociatedMACAddresses": [
    "9c:dc:71:c3:bb:0a",
    "9c:dc:71:c3:bb:0e",
    "9c:dc:71:c3:bb:12",
    "9c:dc:71:c3:bb:16"
  ],
  "FlowControlConfiguration": "None",
  "FlowControlStatus": "None",
  "WakeOnLANEnabled": true,
  "LLDPEnabled": true,
  "LLDPReceive": {
    "ChassisId": "2c:23:3a:48:2f:5d",
    "ChassisIdSubtype": "MacAddr",
    "ManagementAddressIPv4": "",
    "ManagementAddressIPv6": "",
    "ManagementAddressMAC": "2c:23:3a:48:2f:ae",
    "ManagementVlanId": 4095,
    "PortId": "54:65:6E:2D:47:69:67:61:62:69:74:45:74:68:65:72:6E:65:74:31:2F:32:2F:39",
    "PortIdSubtype": "IfName"
  },
  "LLDPTransmit": {
    "ChassisId": "9c:dc:71:c3:bb:16",
    "ChassisIdSubtype": "MacAddr",
    "ManagementAddressIPv4": "",
    "ManagementAddressIPv6": "",
    "ManagementAddressMAC": "9c:dc:71:c3:bb:16",
    "ManagementVlanId": 4095,
    "PortId": "9C:DC:71:C3:BB:16",
    "PortIdSubtype": "MacAddr"
  }
},
"LinkConfiguration": [
  {
    "AutoSpeedNegotiationCapable": true,
    "AutoSpeedNegotiationEnabled": true,
    "CapableLinkSpeedGbps": [
      25.0,
      10.0
    ],
    "ConfiguredNetworkLinks": [
      {
        "ConfiguredLinkSpeedGbps": 25.0,
        "ConfiguredWidth": 1
      },
      {
        "ConfiguredLinkSpeedGbps": 10.0,

```

```

        "ConfiguredWidth": 1
    }
]
},
],
"LinkNetworkTechnology": "Ethernet",
"MaxFrameSize": 9622,
"MaxSpeedGbps": 25.0,
"Width": 1,
"InterfaceEnabled": true,
"SignalDetected": true,
"PortMedium": "Optical",
"LinkState": "Enabled",
"LinkStatus": "LinkUp",
"LinkTransitionIndicator": 1,
"CurrentSpeedGbps": 10.0,
"ActiveWidth": 1,
"SFP": {
    "SupportedSFPTypes": [
        "SFP",
        "SFPPlus",
        "SFP28"
    ],
    "Status": {
        "Health": "OK",
        "State": "Enabled"
    },
    "Manufacturer": "Contoso",
    "PartNumber": "844483-B21",
    "SerialNumber": "THY1020240",
    "MediumType": "FiberOptic",
    "FiberConnectionType": "SingleMode",
    "Type": "SFP28"
},
"@odata.context": "/redfish/v1/$metadata#Port.Port",
"@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports/1",
"@odata.etag": "W/\"B40342B6\"",
"@Redfish.Settings": {
    "@odata.type": "#Settings.v1_3_3.Settings",
    "SettingsObject": {
        "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports/1/Settings"
    },
    "SupportedApplyTimes": [
        "OnReset"
    ]
}
}
}

```



6.4 Get PCIe Information

The Network Adapter is analogous to a PCIe Device, so this next retrieval gets the PCIeDevice information for the link we found in the Network Adapter.

```
GET /redfish/v1/Chassis/1/PCIeDevices/DE07A000
```

The response message contains the following fragment.

```
{
  "@odata.type": "#PCIeDevice.v1_9_0.PCIeDevice",
  "Id": "DE07A000",
  "Name": "PCIe Device",
  "Manufacturer": "Contoso",
  "Model": "Contoso 2",
  "SKU": "Contoso 2 function adapter",
  "SerialNumber": "LMNOP4279",
  "PartNumber": "ABCDEFG2",
  "UUID": "00000000-0000-1000-8000-9cdc71c3bb0a",
  "DeviceType": "MultiFunction",
  "FirmwareVersion": "192.168.59.0",
  "Status": {
    "Health": "OK",
    "State": "Enabled"
  },
  "PCIeFunctions": {
    "@odata.id": "/redfish/v1/Chassis/1/PCIeDevices/DE07A000/PCIeFunctions"
  },
  "PCIeInterface": {
    "LanesInUse": 8,
    "MaxLanes": 16,
    "MaxPCIeType": "Gen4",
    "PCIeType": "Gen4"
  },
  "@odata.context": "/redfish/v1/$metadata#PCIeDevice.PCIeDevice",
  "@odata.id": "/redfish/v1/Chassis/1/PCIeDevices/DE07A000",
  "@odata.etag": "W/\"70C0367C\""
}
```

6.5 Get PCIe Function Information

The PCIeFunction information is linked to the PCIeDevice as well as the NetworkDeviceFunction so we must retrieve that information.


```
GET /redfish/v1/Chassis/1/PCIeDevices/DE07A000/PCIeFunctions/1
```

The response message contains the following fragment.

```
{
  "@odata.type": "#PCIeFunction.v1_3_0.PCIeFunction",
  "Id": "1",
  "Name": "PCIe Function 1",
  "FunctionType": "Physical",
  "DeviceClass": "NetworkController",
  "FunctionId": 1,
  "DeviceId": "0x1801",
  "VendorId": "0x14e4",
  "ClassCode": "0x020000",
  "RevisionId": "0x11",
  "SubsystemId": "0x1598",
  "SubsystemVendorId": "0x14e4",
  "Status": {
    "Health": "OK",
    "State": "Enabled"
  },
  "Links": {
    "NetworkDeviceFunctions": [
      {
        "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/NetworkDeviceFunctions/1"
      }
    ],
    "PCIeDevice": {
      "@odata.id": "/redfish/v1/Chassis/1/PCIeDevices/DE07A000"
    }
  },
  "@odata.context": "/redfish/v1/$metadata#PCIeFunction.PCIeFunction",
  "@odata.id": "/redfish/v1/Chassis/1/PCIeDevices/DE07A000/PCIeFunctions/1",
  "@odata.etag": "W/\"2D186009\""
}
```

6.6 Get System NIC information

EthernetInterface is the System's view of the NIC and will have constructs of what is traditionally the software stack, such as IPv4/IPv6 addresses.

```
GET /redfish/v1/Systems/1/EthernetInterfaces/5
```

The response message contains the following fragment.



```
{
  "@odata.type": "#EthernetInterface.v1_4_1.EthernetInterface",
  "Id": "5",
  "FullDuplex": false,
  "IPv4Addresses": [],
  "IPv4StaticAddresses": [],
  "IPv6AddressPolicyTable": [],
  "IPv6Addresses": [],
  "IPv6StaticAddresses": [],
  "IPv6StaticDefaultGateways": [],
  "InterfaceEnabled": null,
  "LinkStatus": "LinkUp",
  "MACAddress": "9c:dc:71:c3:bb:0a",
  "Name": "",
  "NameServers": [],
  "SpeedMbps": null,
  "StaticNameServers": [],
  "Status": {
    "Health": "OK",
    "State": "Enabled"
  },
  "Links": {
    "NetworkDeviceFunctions": [
      {
        "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/NetworkDeviceFunctions/1"
      }
    ]
  },
  "UefiDevicePath": "PciRoot(0x3)/Pci(0x1,0x1)/Pci(0x0,0x0)",
  "@odata.context": "/redfish/v1/$metadata#EthernetInterface.EthernetInterface",
  "@odata.etag": "W/\"C4D4ADE7\"",
  "@odata.id": "/redfish/v1/Systems/1/EthernetInterfaces/5"
}
```

6.7 Set NIC Information

The previous GETs returned the following properties. By performing a GET operation on that URI, the client can determine what properties can be changed and can set them by performing a PUT or PATCH operation on that resource.

```
"@Redfish.Settings": {
  "@odata.type": "#Settings.v1_3_3.Settings",
  "SettingsObject": {
    "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Settings"
  },
  "SupportedApplyTimes": [
```

```

        "OnReset"
    ]
}

```

6.8 Set Device Function Information

The previous GETs returned the following properties. By performing a GET operation on that URI, the client can determine what properties can be changed and can set them by performing a PUT or PATCH operation on that resource.

```

"@Redfish.Settings": {
  "@odata.type": "#Settings.v1_3_3.Settings",
  "SettingsObject": {
    "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/NetworkDeviceFunctions/1/Settings"
  },
  "SupportedApplyTimes": [
    "OnReset"
  ]
}

```

6.9 Set Port Information

The previous GETs returned the following properties. By performing a GET operation on that URI, the client can determine what properties can be changed and can set them by performing a PUT or PATCH operation on that resource.

```

"@Redfish.Settings": {
  "@odata.type": "#Settings.v1_3_3.Settings",
  "SettingsObject": {
    "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports/1/Settings"
  },
  "SupportedApplyTimes": [
    "OnReset"
  ]
}

```



6.10 Get NIC Metrics

As the NetworkAdapter resource represents the card, or PCIe device, it will have some global metrics that correspond to the device.

```
GET /redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Metrics
```

The response message contains the following fragment.

```
{
  "@odata.type": "#NetworkAdapterMetrics.v1_0_0.NetworkAdapterMetrics",
  "Id": "Metrics",
  "Name": "Network Adapter Metrics",
  "NCSIRXBytes": 20250,
  "NCSIRXFrames": 450,
  "NCSITXBytes": 16168,
  "NCSITXFrames": 450,
  "RXBytes": 1411160,
  "RXUnicastFrames": 9,
  "RXMulticastFrames": 6506,
  "TXBytes": 12179,
  "TXUnicastFrames": 0,
  "TXMulticastFrames": 69,
  "@odata.context": "/redfish/v1/$metadata#NetworkAdapterMetrics.NetworkAdapterMetrics",
  "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Metrics",
  "@odata.etag": "W/\"B85BFB25\""
}
```

6.11 Get Device Function Metrics

For any individual PCIe or virtual function, those metrics can be found on the NetworkDeviceFunctionMetrics.

```
GET /redfish/v1/Chassis/1/NetworkAdapters/DE082000/NetworkDeviceFunctions/0/Metrics
```

The response message contains the following fragment.

```
{
  "@odata.type": "#NetworkDeviceFunctionMetrics.v1_1_0.NetworkDeviceFunctionMetrics",
```

```

    "Id": "Metrics",
    "Name": "Network Device Function 1 Metrics",
    "RXBytes": 286683,
    "RXFrames": 1867,
    "RXUnicastFrames": 0,
    "RXMulticastFrames": 1,
    "TXBytes": 0,
    "TXFrames": 0,
    "TXUnicastFrames": 0,
    "TXMulticastFrames": 0,
    "@odata.context": "/redfish/v1/$metadata#NetworkDeviceFunctionMetrics.NetworkDeviceFunctionMetrics",
    "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/NetworkDeviceFunctions/1/Metrics",
    "@odata.etag": "W/\"7BD1348E\""
  }

```

6.12 Get Port Metrics

Any metrics on the port itself are represented by the PortMetrics object.

```
GET /redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports/1/Metrics
```

The response message contains the following fragment.

```

{
  "@odata.type": "#PortMetrics.v1_2_0.PortMetrics",
  "Id": "Metrics",
  "Name": "Ethernet Port 1 Metrics",
  "Networking": {
    "RXFrames": 8519,
    "RXUnicastFrames": 9,
    "RXMulticastFrames": 6467,
    "RXBroadcastFrames": 2043,
    "RXPFCEFrames": 0,
    "RXDiscards": 0,
    "RXFalseCarrierErrors": 0,
    "RXFCSErrors": 0,
    "RXFrameAlignmentErrors": 0,
    "RXOversizeFrames": 0,
    "RXUndersizeFrames": 0,
    "TXFrames": 69,
    "TXUnicastFrames": 0,
    "TXMulticastFrames": 69,
    "TXBroadcastFrames": 0,
    "TXPFCEFrames": 0,
  }
}

```



```

    "TXDiscards": 0,
    "TXExcessiveCollisions": 0,
    "TXLateCollisions": 0,
    "TXMultipleCollisions": 0,
    "TXSingleCollisions": 0
  },
  "RXBytes": 1401596,
  "RXErrors": 169,
  "TXBytes": 12179,
  "TXErrors": 0,
  "Transceivers": [
    {
      "RXInputPowerMilliWatts": 6985.0,
      "SupplyVoltage": 52096.0,
      "TXBiasCurrentMilliAmps": 3375.0,
      "TXOutputPowerMilliWatts": 7108.0
    }
  ],
  "@odata.context": "/redfish/v1/$metadata#PortMetrics.PortMetrics",
  "@odata.id": "/redfish/v1/Chassis/1/NetworkAdapters/DE07A000/Ports/1/Metrics",
  "@odata.etag": "W/\"4E6105EA\""
}

```

6.13 Set NIC Settings To Default

In order to reset the settings to default, there is a required action called `ResetSettingsToDefault`. This action takes no parameters. It is invoked as follows:

```
POST /redfish/v1/Chassis/1/NetworkAdapter/DE07A000/Actions/NetworkAdapter.ResetSettingsToDefault
```

6.14 Get FRU Information

FRU (Field Replaceable Unit) information is found on the NetworkAdapter. The properties Manufacturer, Model, SKU and SerialNumber are all required properties. For more information, see [GET NetworkAdapter](#).

```
GET /redfish/v1/Chassis/1/NetworkAdapter/DE07A000
```

```
{
```

```

    "Manufacturer": "Contoso",
    "Model": "Contoso 2",
    "SKU": "Contoso 2 function adapter",
    "SerialNumber": "LMNOP4279",
    "PartNumber": "ABCDEFG2"
  }

```

6.15 Get Firmware Version

Like FRU, the firmware version can be found on the NetworkAdapter object under the FirmwarePackageVersion property. This is a required property. For more information, see [GET NetworkAdapter](#).

```
GET /redfish/v1/Chassis/1/NetworkAdapter/DE07A000
```

The response message contains the following fragment.

```

{
  "FirmwarePackageVersion": "1.2.3"
}

```

6.16 Update Firmware

Updating firmware is done via the Redfish UpdateService. The following is an example of how to update firmware on a NetworkAdapter using HTTP multipart. The [Redfish Firmware Update White Paper](#) [6] describes the firmware update process and options.

```

POST /redfish/v1/UpdateService/upload HTTP/1.1
Host: <host-path>
Content-Type: multipart/form-data; boundary=-----e67f97b6546ac967b
Content-Length: <computed-length>
Connection: keep-alive
X-Auth-Token: <session-auth-token>

-----e67f97b6546ac967b
Content-Disposition: form-data; name="UpdateParameters"
Content-Type: application/json

{

```



```
"Targets": ["/redfish/v1/Chassis/1/NetworkAdapter/DE07A000"],
"@Redfish.OperationApplyTime": "OnReset"
}

-----e67f97b6546ac967b
Content-Disposition: form-data; name="UpdateFile"; filename="firmwareimage.bin"
Content-Type: application/octet-stream

<software image binary>
```



7 References

- [1] "[Redfish API Specification](#)"
- [2] "[Redfish Data Model Specification](#)"
- [3] "[Redfish Interop Validator](#)"
- [4] "[OCP Ethernet NIC Profile v1.0](#)"
- [5] "[Redfish Interoperability Profile Specification](#)"
- [6] "[Redfish Firmware Update White Paper](#)"
- [7] "[OCP Ethernet NIC Profile Reference Guide v1.0](#)"
- [8] "[OCP Ethernet NIC mockup](#)"



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